

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.454	125.0	164.0	270.0	7105.294	26.316

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[213,0]	[205,7]	9.899	TIP	BRANCH
2	[117,1]	[205,7]	100.255	TIP	BRANCH
3	[205,7]	[208,17]	10.243	BRANCH	BRANCH
4	[349,5]	[347,16]	10.828	TIP	BRANCH
5	[304,31]	[347,16]	47.799	TIP	BRANCH
6	[347,16]	[380,26]	35.728	BRANCH	BRANCH
7	[441,16]	[449,16]	7.0	BRANCH	BRANCH
8	[432,7]	[441,16]	11.314	TIP	BRANCH
9	[441,16]	[419,51]	43.87	BRANCH	BRANCH
10	[449,16]	[473,70]	63.527	BRANCH	BRANCH
11	[449,16]	[491,20]	43.071	BRANCH	BRANCH
12	[238,0]	[208,17]	36.627	TIP	BRANCH
13	[200,37]	[208,17]	22.314	TIP	BRANCH
14	[66,3]	[57,19]	19.314	TIP	BRANCH
15	[2,32]	[57,19]	80.456	TIP	BRANCH
16	[57,19]	[61,38]	19.657	BRANCH	BRANCH
17	[490,7]	[491,20]	12.414	TIP	BRANCH
18	[491,20]	[473,70]	64.184	BRANCH	BRANCH
19	[381,31]	[380,26]	4.414	TIP	BRANCH
20	[61,38]	[82,48]	24.142	BRANCH	BRANCH
21	[61,38]	[44,90]	63.355	BRANCH	BRANCH
22	[82,48]	[83,60]	11.414	BRANCH	BRANCH
23	[111,0]	[82,48]	60.941	TIP	BRANCH
24	[241,24]	[239,50]	25.828	TIP	BRANCH
25	[270,93]	[239,50]	57.184	TIP	BRANCH
26	[216,78]	[239,50]	38.284	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[419,51]	[455,75]	44.941	BRANCH	BRANCH
28	[419,51]	[395,56]	25.071	BRANCH	BRANCH
29	[10,40]	[15,54]	15.071	TIP	BRANCH
30	[18,54]	[15,54]	2.0	TIP	BRANCH
31	[9,56]	[15,54]	5.828	TIP	BRANCH
32	[395,56]	[394,72]	15.414	BRANCH	BRANCH
33	[369,57]	[395,56]	28.314	TIP	BRANCH
34	[83,60]	[111,78]	34.456	BRANCH	BRANCH
35	[83,60]	[44,90]	52.184	BRANCH	BRANCH
36	[473,70]	[471,75]	4.828	BRANCH	BRANCH
37	[366,76]	[394,72]	28.657	TIP	BRANCH
38	[394,72]	[444,88]	56.799	BRANCH	BRANCH
39	[455,75]	[471,75]	15.0	BRANCH	BRANCH
40	[455,75]	[444,88]	16.142	BRANCH	BRANCH
41	[471,75]	[483,87]	17.142	BRANCH	BRANCH
42	[111,78]	[132,80]	21.657	BRANCH	BRANCH
43	[110,84]	[111,78]	5.414	TIP	BRANCH
44	[167,78]	[132,80]	35.657	TIP	BRANCH
45	[131,82]	[132,80]	1.414	TIP	BRANCH
46	[483,87]	[487,127]	41.243	BRANCH	BRANCH
47	[501,82]	[483,87]	18.657	TIP	BRANCH
48	[444,88]	[446,109]	20.414	BRANCH	BRANCH
49	[44,90]	[16,119]	42.527	BRANCH	BRANCH
50	[378,90]	[378,103]	12.0	TIP	BRANCH
51	[392,108]	[378,103]	15.071	TIP	BRANCH
52	[328,109]	[378,103]	70.941	TIP	BRANCH
53	[446,109]	[429,136]	33.627	BRANCH	BRANCH
54	[453,115]	[446,109]	8.071	TIP	BRANCH
55	[0,95]	[16,119]	31.385	TIP	BRANCH
56	[16,119]	[54,139]	46.456	BRANCH	BRANCH
57	[502,120]	[487,127]	17.728	TIP	BRANCH
58	[487,127]	[482,142]	16.071	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[225,128]	[267,128]	44.071	TIP	BRANCH
60	[272,127]	[267,128]	4.414	TIP	BRANCH
61	[232,166]	[267,128]	53.841	TIP	BRANCH
62	[399,126]	[425,136]	31.385	TIP	BRANCH
63	[425,136]	[429,136]	3.0	BRANCH	BRANCH
64	[425,136]	[411,162]	30.385	BRANCH	BRANCH
65	[429,136]	[451,142]	23.657	BRANCH	BRANCH
66	[81,133]	[54,139]	29.314	TIP	BRANCH
67	[54,139]	[36,275]	171.711	BRANCH	BRANCH
68	[55,121]	[54,139]	17.828	TIP	BRANCH
69	[451,142]	[477,143]	25.414	BRANCH	BRANCH
70	[451,142]	[433,179]	43.042	BRANCH	BRANCH
71	[482,142]	[477,143]	4.414	BRANCH	BRANCH
72	[496,148]	[482,142]	15.899	TIP	BRANCH
73	[477,143]	[472,177]	35.071	BRANCH	BRANCH
74	[355,146]	[334,168]	34.971	BRANCH	BRANCH
75	[355,146]	[338,179]	43.042	BRANCH	BRANCH
76	[401,141]	[355,146]	52.456	TIP	BRANCH
77	[411,162]	[392,175]	23.971	BRANCH	BRANCH
78	[411,162]	[412,168]	5.0	BRANCH	BRANCH
79	[299,153]	[334,168]	41.042	TIP	BRANCH
80	[334,168]	[338,179]	11.657	BRANCH	BRANCH
81	[412,168]	[416,177]	10.243	BRANCH	BRANCH
82	[412,168]	[410,169]	1.414	BRANCH	BRANCH
83	[410,169]	[409,171]	2.828	BRANCH	PATH
84	[383,175]	[392,175]	8.0	BRANCH	BRANCH
85	[377,167]	[383,175]	9.071	TIP	BRANCH
86	[383,175]	[367,211]	42.385	BRANCH	BRANCH
87	[392,175]	[401,183]	11.899	BRANCH	BRANCH
88	[416,177]	[433,179]	16.828	BRANCH	BRANCH
89	[416,177]	[401,183]	16.071	BRANCH	BRANCH
90	[472,177]	[499,219]	61.213	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[472,177]	[435,187]	40.142	BRANCH	BRANCH
92	[338,179]	[332,210]	32.899	BRANCH	BRANCH
93	[433,179]	[435,187]	7.828	BRANCH	BRANCH
94	[401,183]	[401,204]	20.0	BRANCH	BRANCH
95	[435,187]	[435,189]	1.0	BRANCH	BRANCH
96	[435,189]	[462,226]	47.77	BRANCH	BRANCH
97	[435,189]	[409,206]	33.213	BRANCH	BRANCH
98	[27,166]	[11,190]	29.627	TIP	BRANCH
99	[8,190]	[11,190]	2.0	TIP	BRANCH
100	[11,190]	[36,275]	93.941	BRANCH	BRANCH
101	[401,204]	[409,206]	7.828	BRANCH	BRANCH
102	[401,204]	[378,222]	31.213	BRANCH	BRANCH
103	[409,206]	[409,217]	10.0	BRANCH	BRANCH
104	[332,210]	[354,214]	22.657	BRANCH	BRANCH
105	[332,210]	[322,218]	11.899	BRANCH	BRANCH
106	[367,211]	[354,214]	13.243	BRANCH	BRANCH
107	[367,211]	[378,222]	14.728	BRANCH	BRANCH
108	[354,221]	[354,214]	6.828	TIP	BRANCH
109	[409,217]	[378,225]	35.799	BRANCH	BRANCH
110	[409,217]	[421,238]	24.556	BRANCH	BRANCH
111	[256,225]	[322,218]	75.355	TIP	BRANCH
112	[322,218]	[303,249]	38.456	BRANCH	BRANCH
113	[499,219]	[464,226]	36.899	BRANCH	BRANCH
114	[507,228]	[499,219]	10.899	TIP	BRANCH
115	[378,222]	[378,225]	2.0	BRANCH	BRANCH
116	[378,225]	[373,269]	48.799	BRANCH	BRANCH
117	[462,226]	[464,226]	1.0	BRANCH	BRANCH
118	[462,226]	[421,238]	44.556	BRANCH	BRANCH
119	[464,226]	[453,258]	35.556	BRANCH	BRANCH
120	[421,238]	[403,259]	28.213	BRANCH	BRANCH
121	[122,232]	[126,240]	8.657	TIP	BRANCH
122	[129,240]	[126,240]	2.0	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[125,242]	[126,240]	1.414	TIP	BRANCH
124	[148,245]	[152,248]	4.243	TIP	BRANCH
125	[113,274]	[152,248]	56.87	TIP	BRANCH
126	[172,199]	[152,248]	55.87	TIP	BRANCH
127	[274,248]	[303,249]	29.243	TIP	BRANCH
128	[303,249]	[304,263]	13.414	BRANCH	BRANCH
129	[453,258]	[454,263]	4.414	BRANCH	BRANCH
130	[445,255]	[453,258]	8.657	TIP	BRANCH
131	[403,259]	[373,269]	33.142	BRANCH	BRANCH
132	[403,259]	[403,275]	15.0	BRANCH	BRANCH
133	[304,263]	[351,284]	55.87	BRANCH	BRANCH
134	[304,263]	[291,285]	25.971	BRANCH	BRANCH
135	[454,263]	[454,274]	140.569	BRANCH	BRANCH
136	[454,263]	[454,274]	10.0	BRANCH	BRANCH
137	[373,269]	[351,284]	29.556	BRANCH	BRANCH
138	[454,274]	[420,284]	36.728	BRANCH	BRANCH
139	[33,308]	[36,275]	34.071	TIP	BRANCH
140	[231,252]	[234,275]	23.243	TIP	BRANCH
141	[215,278]	[234,275]	19.243	TIP	BRANCH
142	[247,287]	[234,275]	16.556	TIP	BRANCH
143	[403,275]	[420,284]	20.899	BRANCH	BRANCH
144	[403,275]	[402,310]	41.385	BRANCH	BRANCH
145	[330,315]	[351,284]	43.971	TIP	BRANCH
146	[420,284]	[402,310]	32.042	BRANCH	BRANCH
147	[263,290]	[291,285]	30.728	TIP	BRANCH
148	[291,285]	[303,288]	12.828	BRANCH	BRANCH
149	[310,282]	[303,288]	9.071	TIP	BRANCH
150	[303,288]	[305,302]	13.828	BRANCH	BRANCH
151	[311,303]	[305,302]	5.414	TIP	BRANCH
152	[300,306]	[305,302]	5.243	TIP	BRANCH
153	[402,310]	[399,324]	14.243	BRANCH	BRANCH
154	[454,296]	[451,310]	14.243	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[451,310]	[433,323]	24.728	BRANCH	BRANCH
156	[451,310]	[468,322]	20.556	BRANCH	BRANCH
157	[69,319]	[42,349]	43.698	BRANCH	BRANCH
158	[64,314]	[69,319]	5.657	TIP	BRANCH
159	[93,304]	[69,319]	29.385	TIP	BRANCH
160	[291,321]	[293,321]	1.0	BRANCH	BRANCH
161	[291,321]	[288,361]	41.071	BRANCH	BRANCH
162	[291,321]	[222,386]	108.811	BRANCH	BRANCH
163	[296,315]	[293,321]	6.243	TIP	BRANCH
164	[295,322]	[293,321]	1.0	TIP	BRANCH
165	[468,322]	[468,327]	4.0	BRANCH	BRANCH
166	[468,322]	[490,342]	40.728	BRANCH	BRANCH
167	[433,323]	[402,324]	30.414	BRANCH	BRANCH
168	[433,323]	[456,332]	25.314	BRANCH	BRANCH
169	[399,324]	[402,324]	2.0	BRANCH	BRANCH
170	[399,324]	[369,342]	37.627	BRANCH	BRANCH
171	[402,324]	[403,344]	19.414	BRANCH	BRANCH
172	[468,327]	[486,342]	23.799	BRANCH	BRANCH
173	[468,327]	[456,332]	13.071	BRANCH	BRANCH
174	[456,332]	[442,351]	26.142	BRANCH	BRANCH
175	[369,342]	[335,347]	35.071	BRANCH	BRANCH
176	[369,342]	[369,351]	8.0	BRANCH	BRANCH
177	[486,342]	[490,342]	3.0	BRANCH	BRANCH
178	[465,360]	[486,342]	28.213	TIP	BRANCH
179	[490,342]	[498,343]	7.414	BRANCH	BRANCH
180	[503,340]	[498,343]	4.828	TIP	BRANCH
181	[508,366]	[498,343]	25.728	TIP	BRANCH
182	[403,344]	[442,351]	42.556	BRANCH	BRANCH
183	[403,344]	[369,351]	38.971	BRANCH	BRANCH
184	[323,346]	[335,347]	11.828	TIP	BRANCH
185	[335,347]	[288,361]	55.728	BRANCH	BRANCH
186	[6,356]	[42,349]	38.485	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[67,374]	[42,349]	39.627	TIP	BRANCH
188	[369,351]	[359,377]	28.728	BRANCH	BRANCH
189	[442,351]	[459,394]	49.042	BRANCH	BRANCH
190	[264,361]	[288,361]	24.657	TIP	BRANCH
191	[359,377]	[337,407]	41.042	BRANCH	BRANCH
192	[354,374]	[359,377]	4.828	TIP	BRANCH
193	[222,386]	[253,439]	74.799	BRANCH	BRANCH
194	[205,376]	[222,386]	19.728	TIP	BRANCH
195	[424,382]	[416,390]	10.899	TIP	BRANCH
196	[386,397]	[416,390]	31.899	TIP	BRANCH
197	[416,390]	[422,413]	24.071	BRANCH	BRANCH
198	[489,378]	[494,391]	14.071	TIP	BRANCH
199	[508,396]	[494,391]	15.071	TIP	BRANCH
200	[494,391]	[481,407]	20.971	BRANCH	BRANCH
201	[459,394]	[481,407]	26.385	BRANCH	BRANCH
202	[451,399]	[459,394]	8.657	TIP	BRANCH
203	[315,398]	[314,400]	1.414	TIP	BRANCH
204	[312,399]	[314,400]	1.414	TIP	BRANCH
205	[314,400]	[315,404]	3.414	BRANCH	BRANCH
206	[315,404]	[337,407]	22.243	BRANCH	BRANCH
207	[315,404]	[312,432]	29.071	BRANCH	BRANCH
208	[337,407]	[341,429]	22.657	BRANCH	BRANCH
209	[481,407]	[491,443]	39.142	BRANCH	BRANCH
210	[448,422]	[422,413]	28.728	TIP	BRANCH
211	[422,413]	[397,422]	27.314	BRANCH	BRANCH
212	[511,406]	[507,414]	8.243	TIP	BRANCH
213	[505,415]	[507,414]	1.0	TIP	BRANCH
214	[510,418]	[507,414]	3.828	TIP	BRANCH
215	[397,422]	[376,426]	21.657	BRANCH	BRANCH
216	[397,422]	[387,443]	25.899	BRANCH	BRANCH
217	[376,426]	[341,429]	35.243	BRANCH	BRANCH
218	[376,426]	[387,443]	20.556	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[358,495]	[341,429]	72.042	TIP	BRANCH
220	[341,429]	[312,432]	28.828	BRANCH	BRANCH
221	[312,432]	[310,433]	1.414	BRANCH	BRANCH
222	[310,433]	[277,434]	32.414	BRANCH	BRANCH
223	[308,452]	[310,433]	18.828	TIP	BRANCH
224	[293,389]	[277,434]	68.042	TIP	BRANCH
225	[277,434]	[253,439]	24.657	BRANCH	BRANCH
226	[253,454]	[253,439]	14.828	TIP	BRANCH
227	[387,443]	[385,454]	10.828	BRANCH	BRANCH
228	[477,440]	[491,443]	14.243	TIP	BRANCH
229	[508,453]	[491,443]	20.314	TIP	BRANCH
230	[366,450]	[385,454]	20.485	TIP	BRANCH
231	[385,454]	[482,485]	135.083	BRANCH	BRANCH
232	[1,467]	[5,480]	13.657	TIP	BRANCH
233	[78,495]	[5,480]	79.87	TIP	BRANCH
234	[1,487]	[5,480]	7.243	TIP	BRANCH
235	[507,486]	[482,485]	27.728	TIP	BRANCH
236	[482,485]	[460,506]	33.799	BRANCH	BRANCH
237	[458,502]	[460,506]	3.828	TIP	BRANCH
238	[454,510]	[460,506]	6.243	TIP	BRANCH
239	[385,2]	[387,12]	11.657	TIP	TIP
240	[281,47]	[284,55]	11.243	TIP	TIP
241	[303,47]	[302,48]	1.414	TIP	TIP
242	[358,54]	[338,72]	29.213	TIP	TIP
243	[191,60]	[179,75]	22.314	TIP	TIP
244	[208,97]	[135,138]	114.569	TIP	TIP
245	[281,146]	[287,150]	7.657	TIP	TIP
246	[121,155]	[130,167]	20.071	TIP	TIP
247	[108,156]	[89,209]	63.941	TIP	TIP
248	[165,157]	[184,176]	30.385	TIP	TIP
249	[273,171]	[242,181]	35.142	TIP	TIP
250	[183,183]	[177,191]	11.071	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
251	[203,206]	[191,216]	16.142	TIP	TIP
252	[206,234]	[211,241]	10.243	TIP	TIP
253	[1,270]	[13,276]	15.071	TIP	TIP
254	[209,298]	[224,318]	31.728	TIP	TIP
255	[62,300]	[61,303]	3.414	TIP	TIP
256	[263,304]	[260,307]	4.243	TIP	TIP
257	[11,307]	[0,317]	16.314	TIP	TIP
258	[176,322]	[138,341]	45.87	TIP	TIP
259	[215,343]	[212,345]	3.828	TIP	TIP
260	[104,344]	[102,353]	9.828	TIP	TIP
261	[330,380]	[327,384]	5.243	TIP	TIP
262	[50,383]	[57,397]	16.899	TIP	TIP
263	[105,389]	[108,389]	3.828	TIP	TIP
264	[189,413]	[193,417]	5.657	TIP	TIP
265	[15,421]	[17,438]	19.485	TIP	TIP
266	[416,445]	[425,454]	13.314	TIP	TIP
267	[90,449]	[94,453]	5.657	TIP	TIP
268	[269,467]	[266,470]	4.243	TIP	TIP
269	[233,475]	[230,476]	3.414	TIP	TIP
270	[423,498]	[435,511]	19.728	TIP	TIP